

CYBERSECURITY INTERNSHIP PROJECT REPORT

Authentication Security Assessment

2FA Bypass Using a Brute-Force Attack

Web Authentication & Multi-Factor Authentication Security Assessment

Platform: PortSwigger Web Security Academy

TOOLS USED

Burp Suite Professional • Firefox Browser • Kali Linux • Oracle VirtualBox • PortSwigger Web Security Academy

Prepared By	MANOJ
Intern ID	CA0209
Internship Organization	Cryptonic Area Cybersecurity Internship Program
Report Type	Authentication Security Assessment Report
Assessment Category	Authentication Security Testing
Lab Name	2FA Bypass Using a Brute-Force Attack
Difficulty Level	Expert
Platform	PortSwigger Web Security Academy
Testing Environment	Kali Linux with Burp Suite Professional
Submission Date	31 July 2026

Table of Contents

1. Introduction
2. Objectives
3. Lab Information
4. Tools Used
5. Methodology
6. Practical Implementation
7. Results
8. Security Recommendations
9. Skills Learned
10. Conclusion

1. Introduction

Web authentication verifies a user's identity before granting access to protected resources. Modern applications commonly implement Multi-Factor Authentication (MFA), including Two-Factor Authentication (2FA), to reduce the impact of stolen credentials. Authentication remains a primary target for attackers through password guessing, credential stuffing, brute-force attacks, session hijacking, insecure session management, and flawed authentication logic. The PortSwigger Web Security Academy provides realistic, intentionally vulnerable laboratories that allow security professionals to safely practice identifying and validating authentication weaknesses. This assessment focuses on a controlled brute-force attack against a vulnerable 2FA implementation using Burp Suite Professional, session handling rules, macros, and Intruder. All testing was performed only within the authorized training environment.

2. Objectives

- Understand authentication mechanisms.
- Analyze 2FA implementation.
- Learn Burp Suite Professional workflow.
- Configure session handling rules.
- Configure Burp Macros.
- Automate authentication.
- Perform controlled brute-force testing in a lab environment.
- Identify successful authentication responses.

3. Lab Information

Lab Name	2FA Bypass Using a Brute Force Attack
Difficulty	Expert
Platform	PortSwigger Web Security Academy
Category	Authentication
Application	Deliberately vulnerable training lab

4. Tools Used

Tool	Purpose
Burp Suite Professional	Intercepting, macros, Intruder and session handling
Firefox	Target access
Kali Linux	Testing OS
Oracle VirtualBox	Virtual lab
PortSwigger Web Security Academy	Training platform

5. Methodology

Information Gathering → Authentication Analysis → Request Inspection → Session Analysis → Macro Creation → Session Handling Rules → Intruder Configuration → Payload Testing → Response Analysis → Successful Authentication.

6. Practical Implementation

Figure 1: Authentication Labs Overview

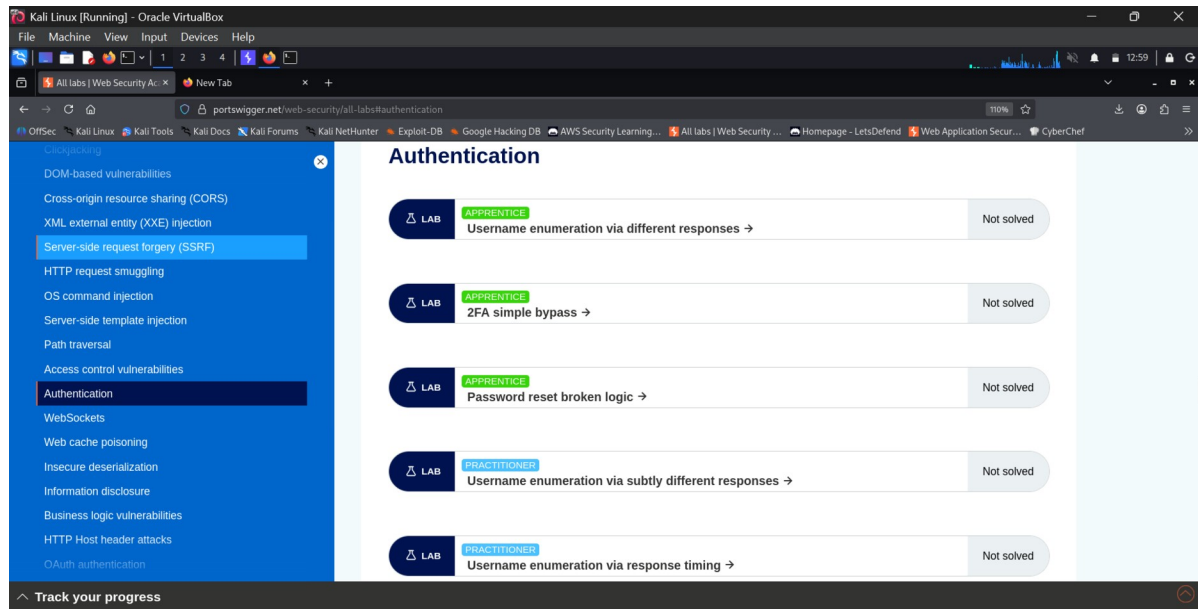


Figure 1: Authentication Labs Overview

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 2: Lab Overview

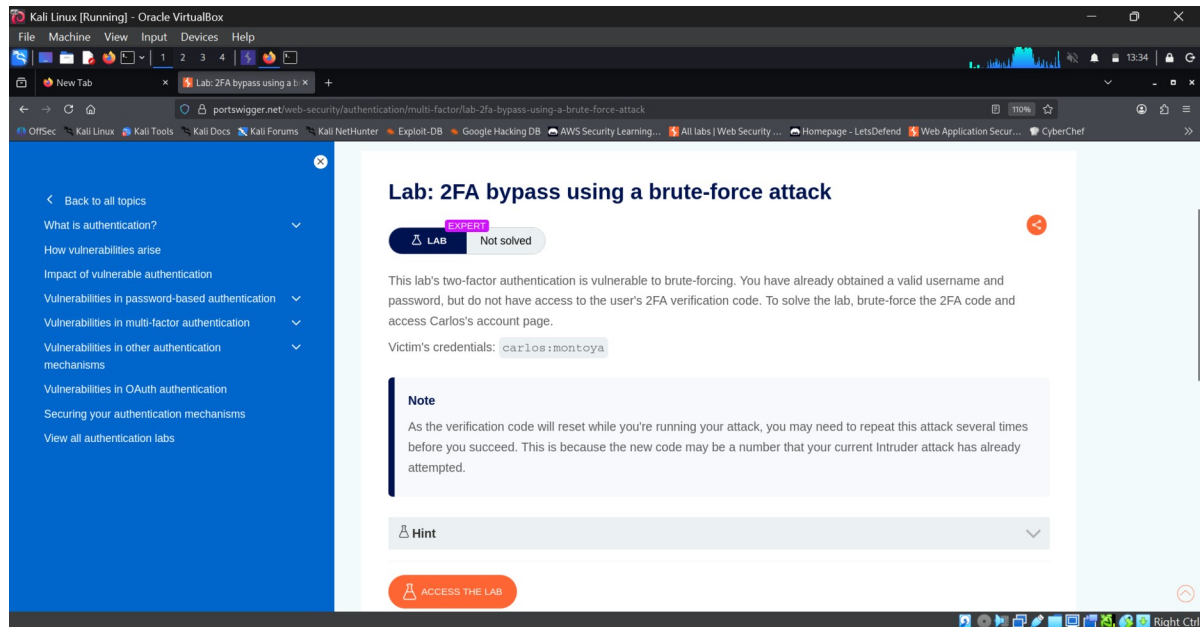


Figure 2: Lab Overview

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 3: Burp Suite Professional Environment

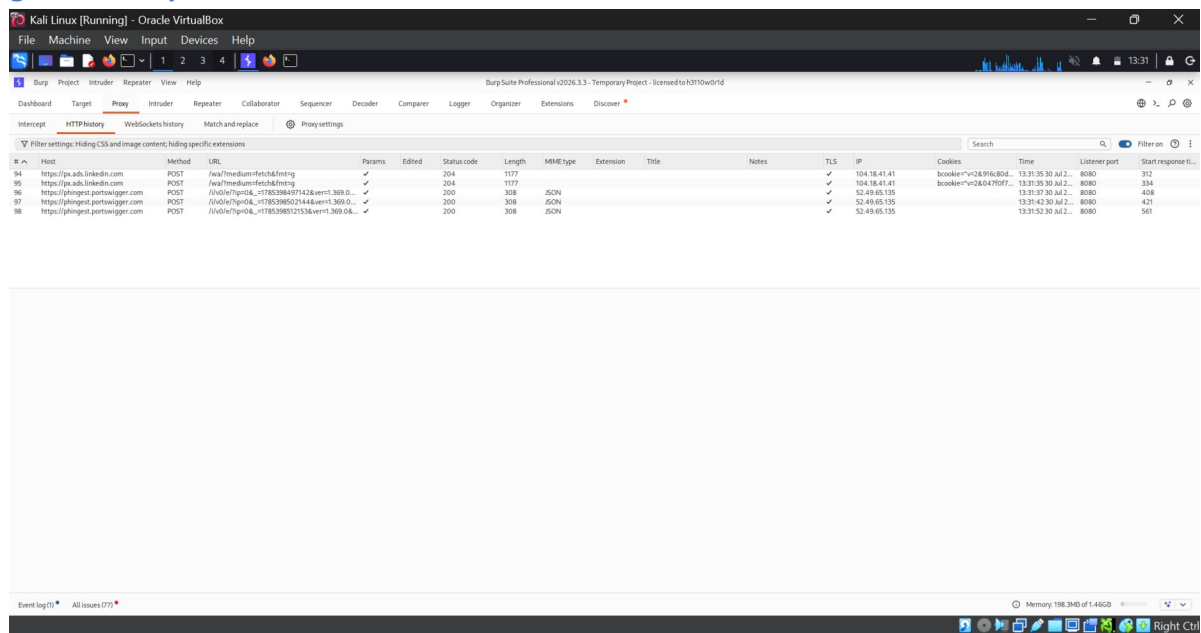


Figure 3: Burp Suite Professional Environment

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 4: Target Login Page

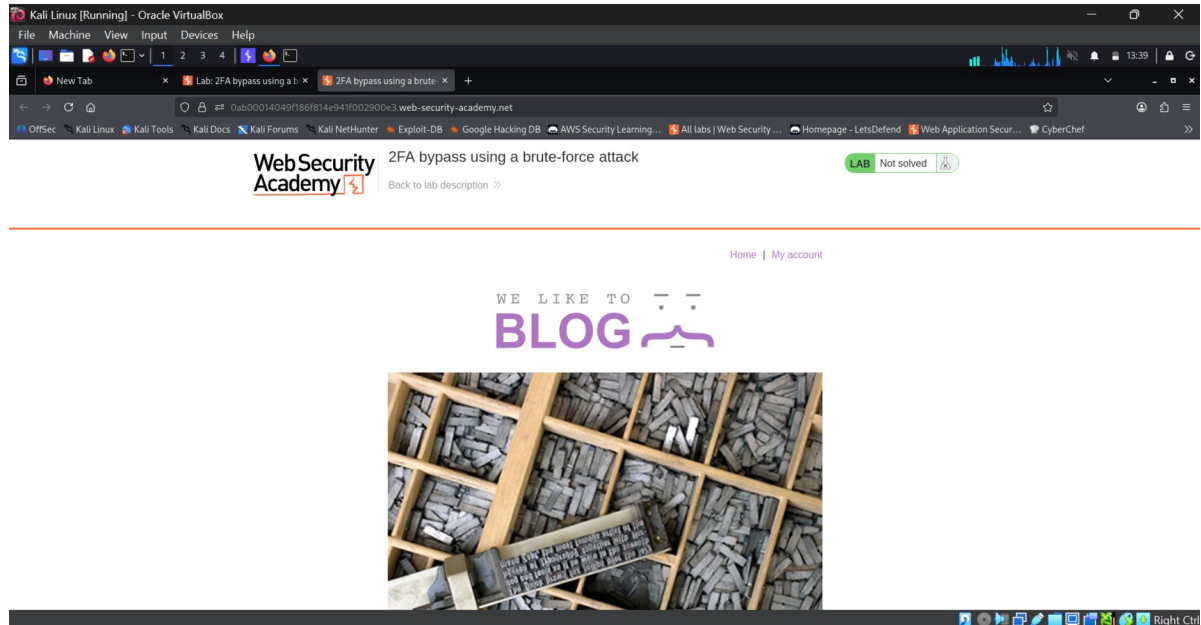


Figure 4: Target Login Page

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 5: Login Using Valid Credentials

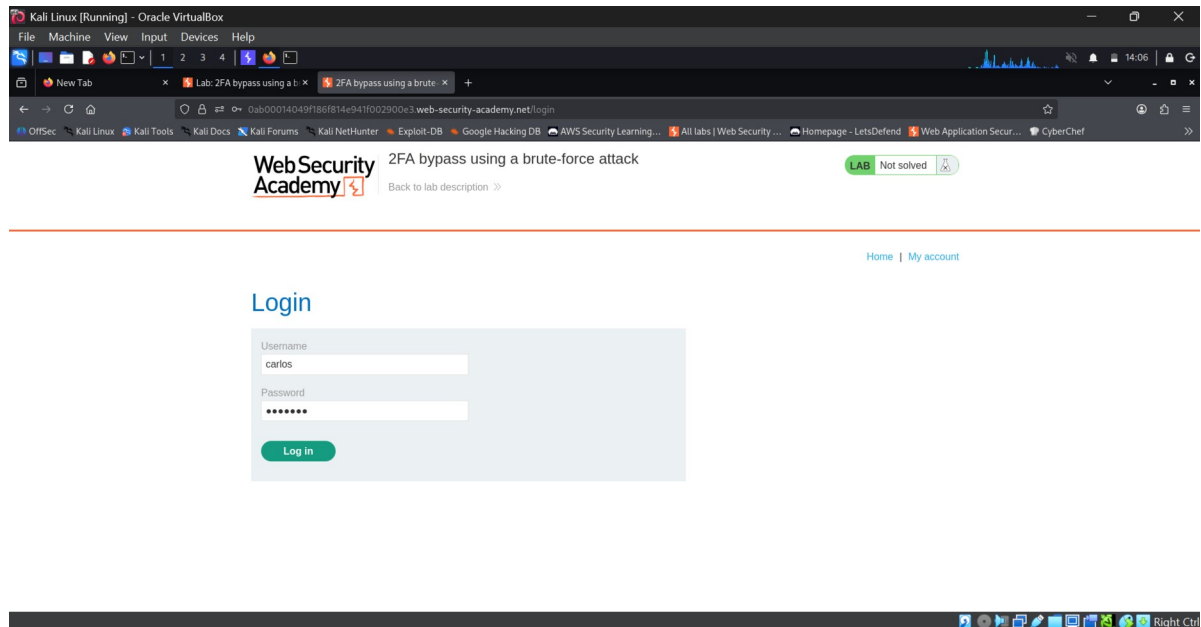


Figure 5: Login Using Valid Credentials

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 6: Two Factor Authentication Page

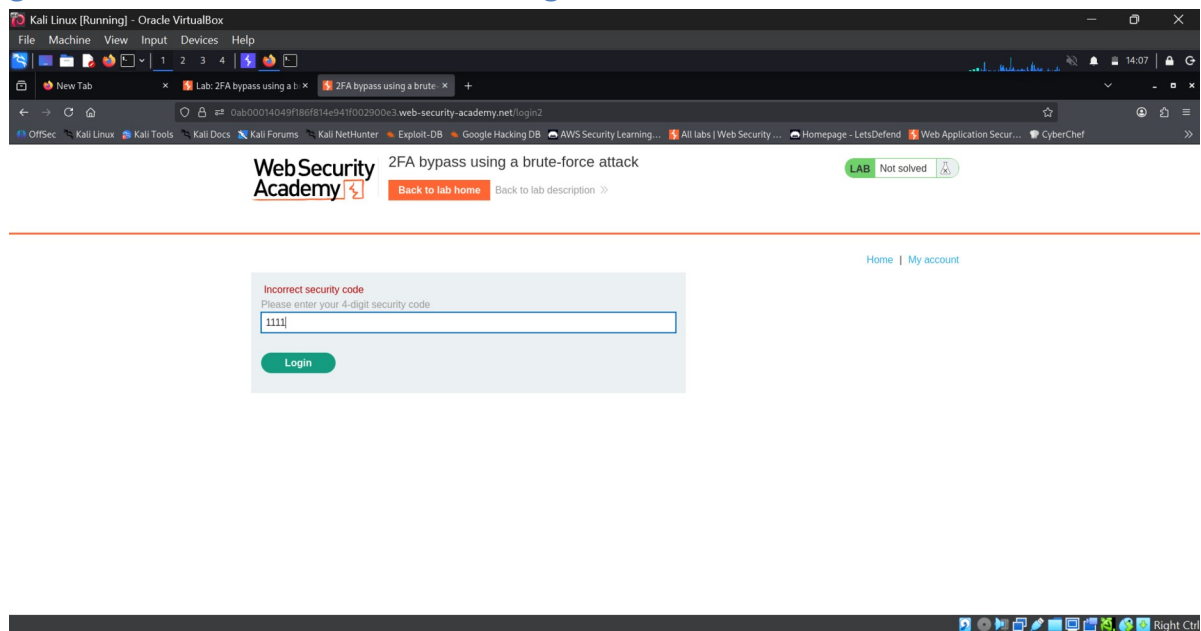


Figure 6: Two Factor Authentication Page

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 7: HTTP Request Captured

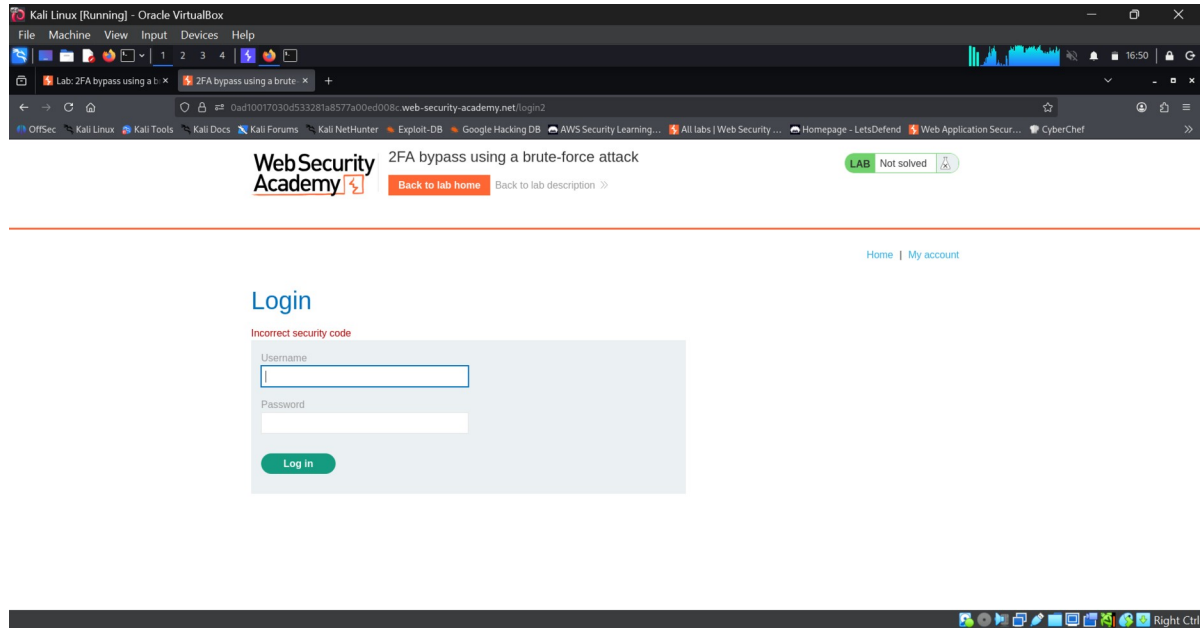


Figure 7: HTTP Request Captured

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 8: CSRF Token Inspection

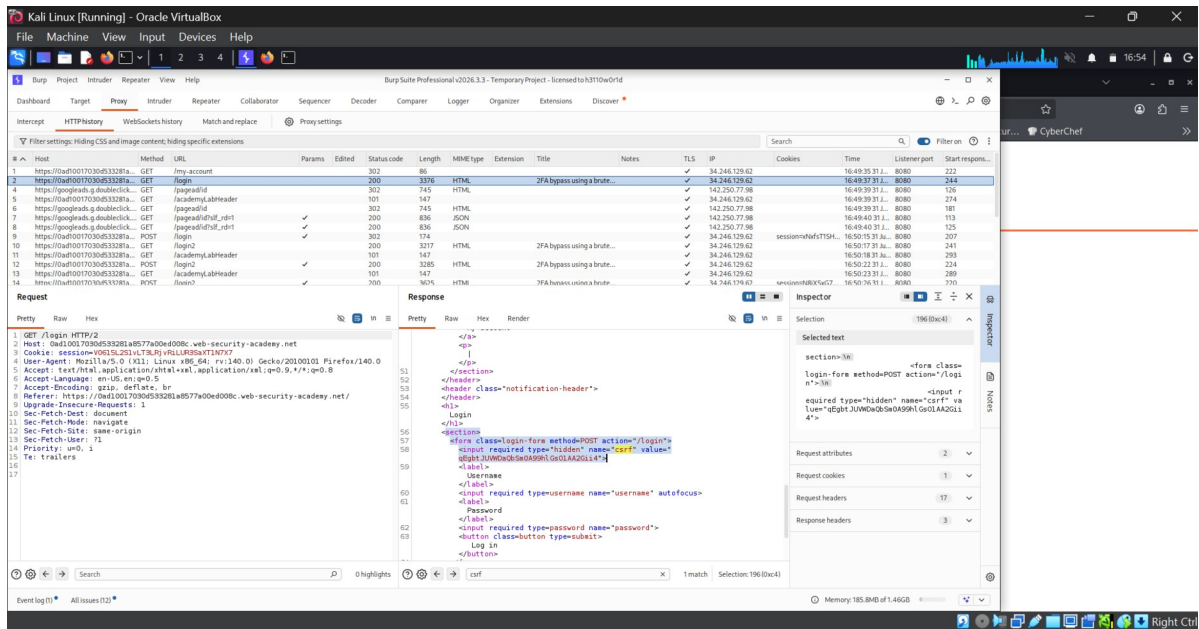


Figure 8: CSRF Token Inspection

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 9: Session Handling Rule Configuration

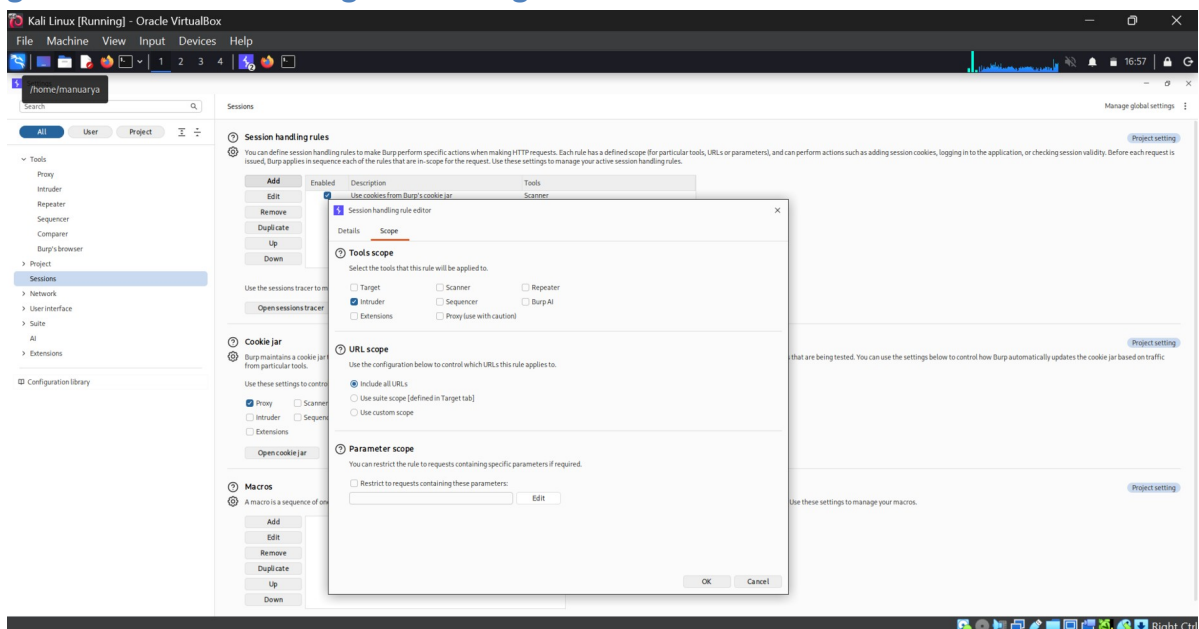


Figure 9: Session Handling Rule Configuration

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 10: Macro Recording

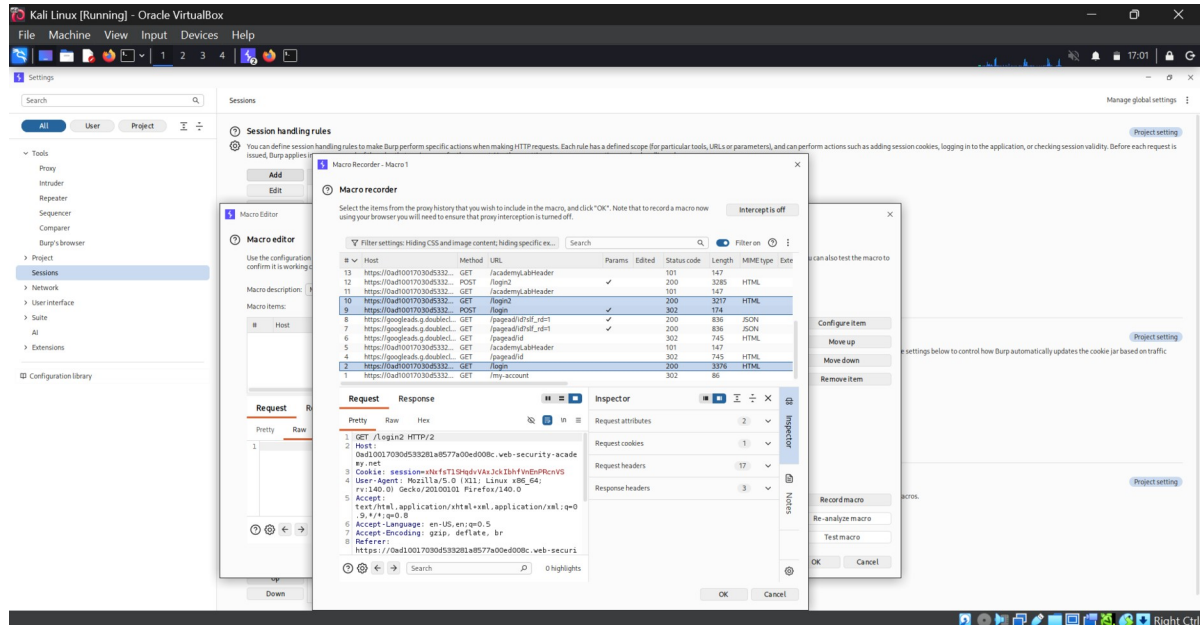


Figure 10: Macro Recording

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

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Figure 13: Intruder Payload Configuration

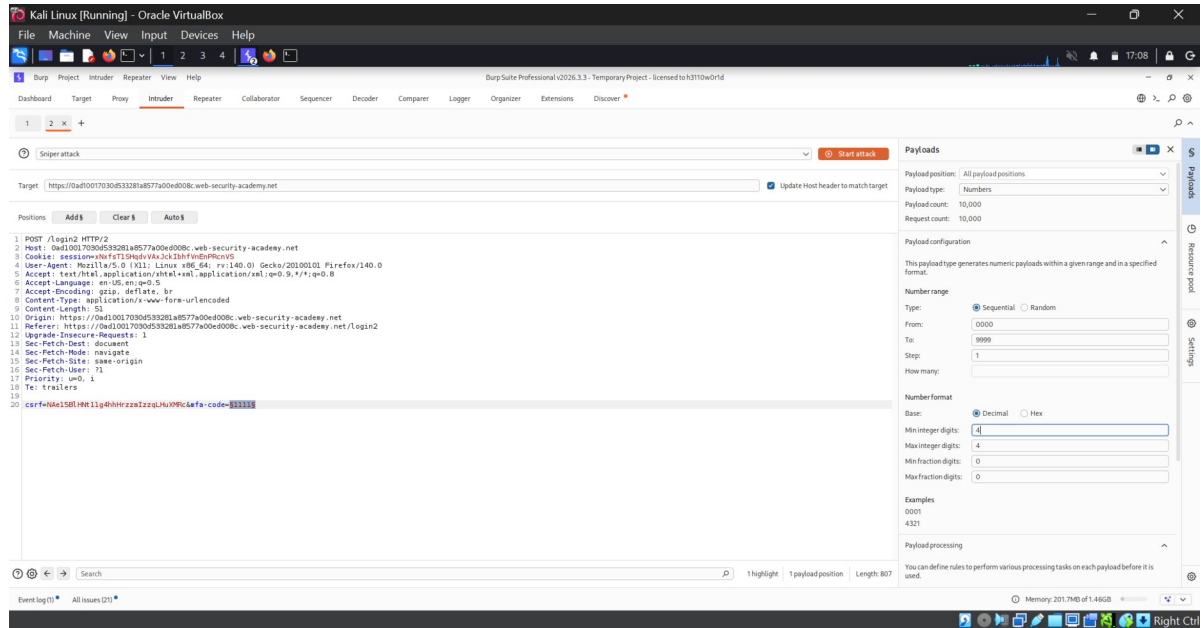


Figure 13: Intruder Payload Configuration

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 14: Intruder Attack Results

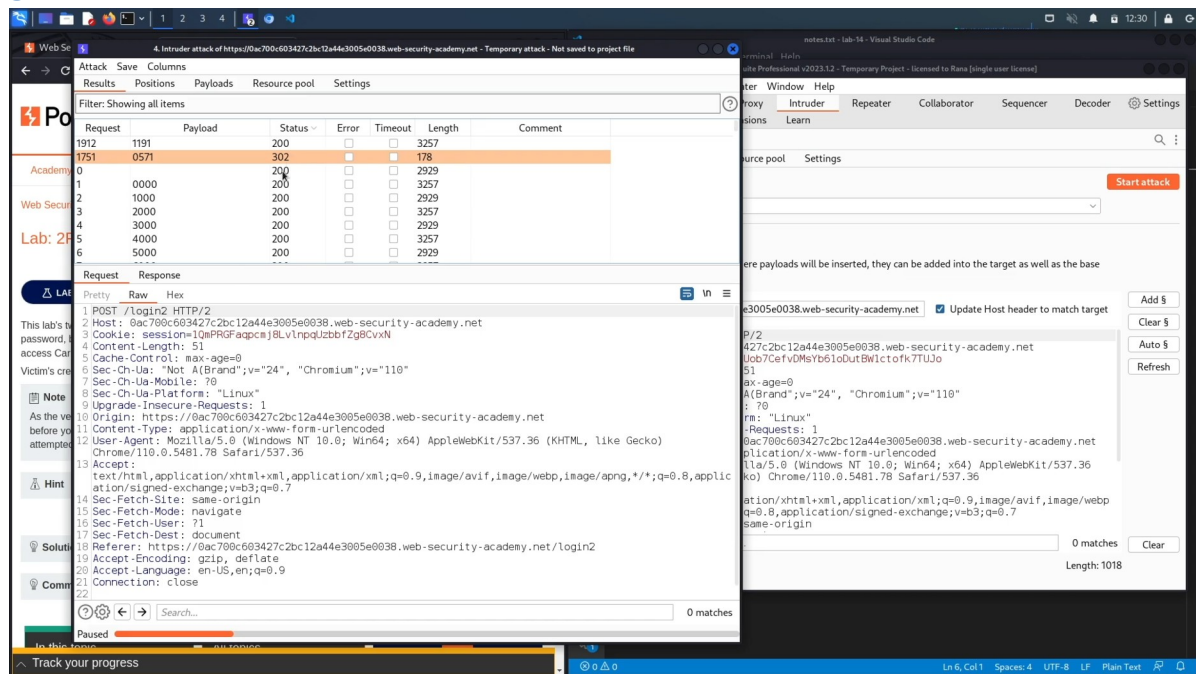


Figure 14: Intruder Attack Results

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 15: Successful HTTP 302 Response

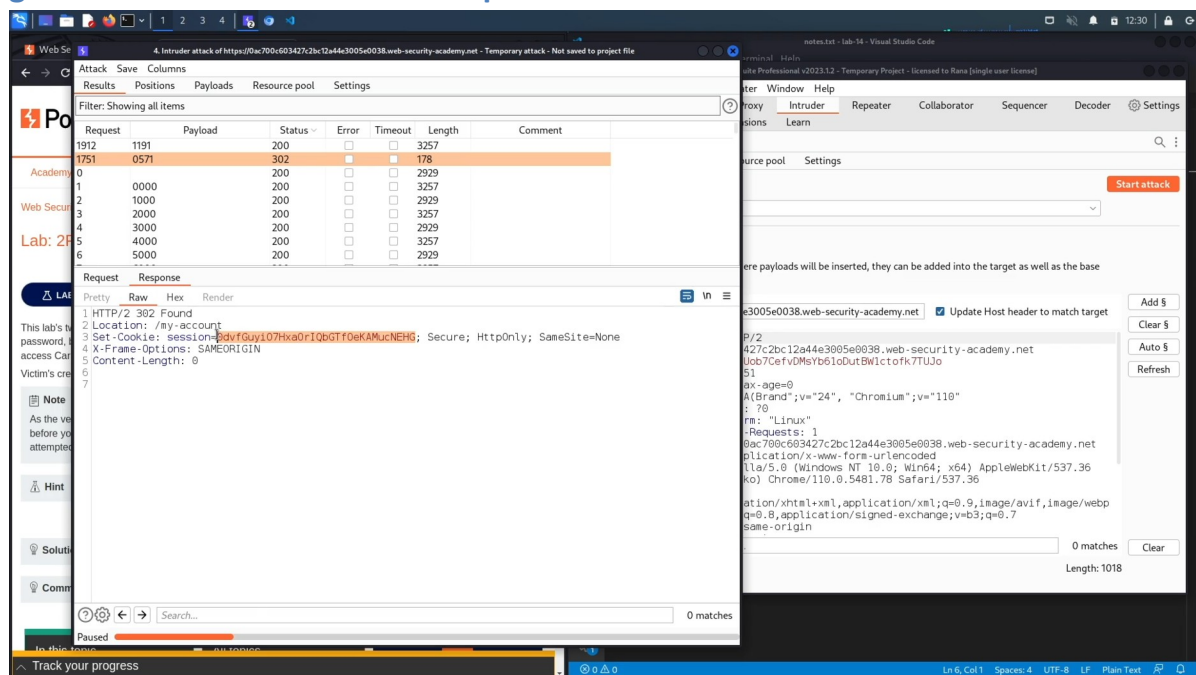


Figure 15: Successful HTTP 302 Response

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 16: Authenticated Browser Session

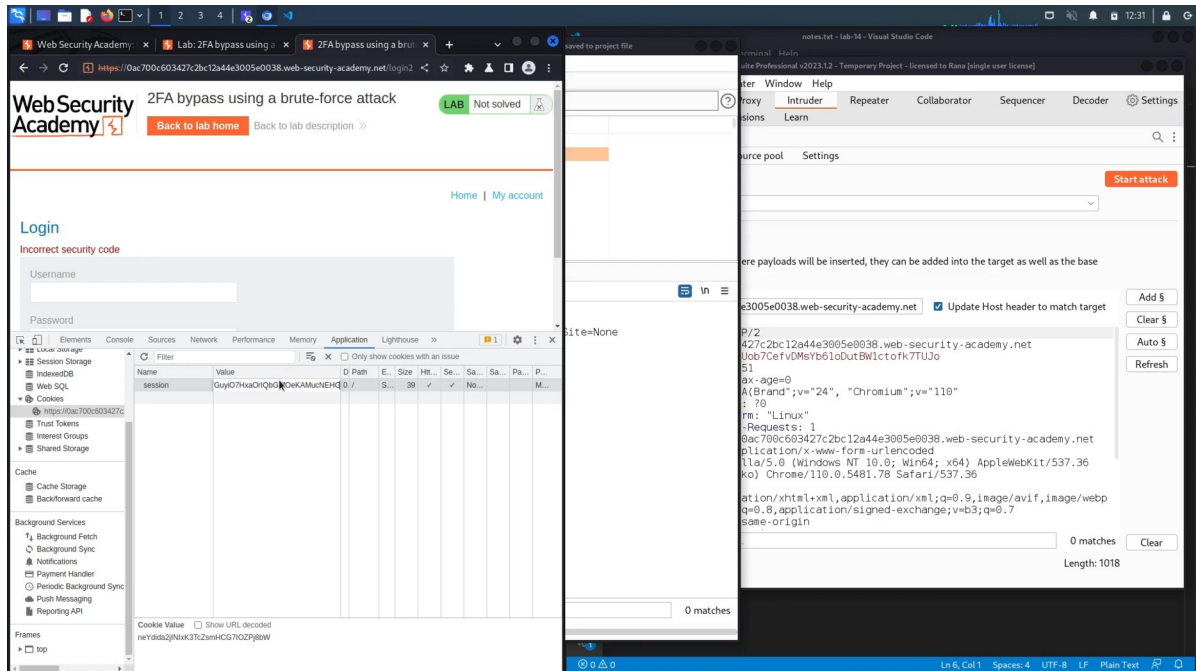


Figure 16: Authenticated Browser Session

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

Figure 17: Lab Successfully Completed

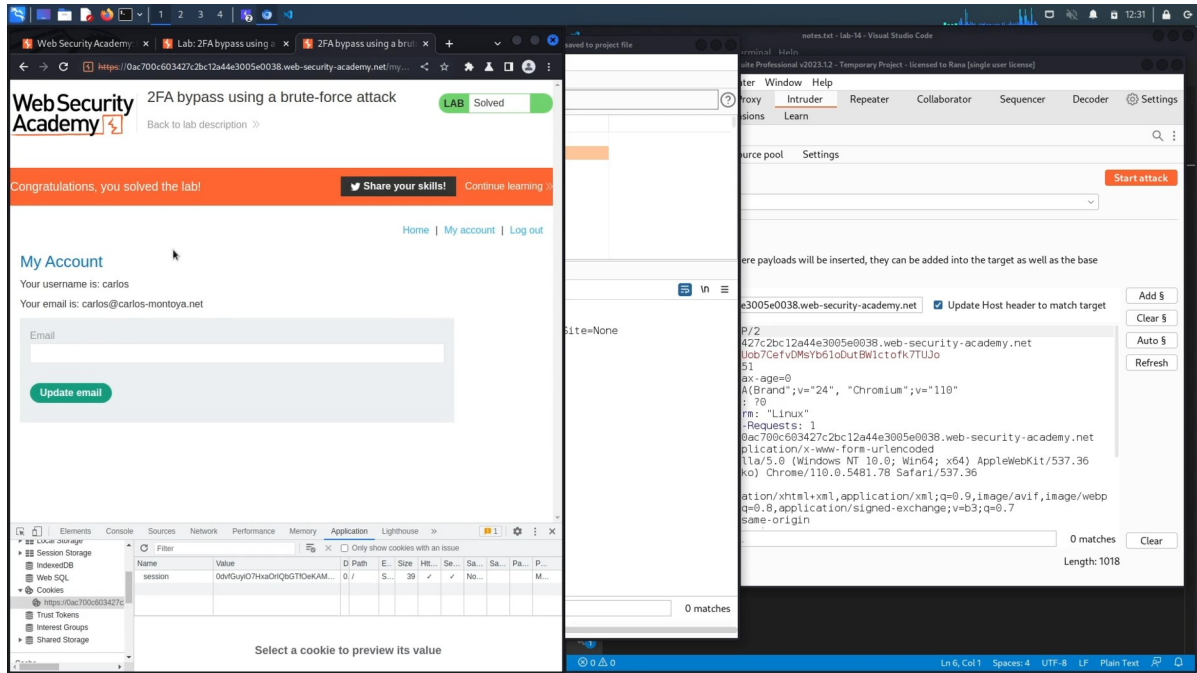


Figure 17: Lab Successfully Completed

Description: This step demonstrates the testing workflow, why it was required, the actions performed, the observed behavior, expected outcome, and the associated authentication security concept within the controlled PortSwigger lab.

7. Results

Objective	Status
Authentication Analyzed	Completed
Session Handling	Completed
Macro Configuration	Completed
Intruder Configuration	Completed
Payload Testing	Completed
Successful Authentication	Completed
Expert Lab Completed	Completed

8. Security Recommendations

- Increase OTP length
- Rate limiting
- Temporary account lockout
- CAPTCHA
- Adaptive authentication
- Device fingerprinting

- Risk-based authentication
- Strong session management
- Monitoring
- Logging

9. Skills Learned

- Burp Suite Professional
- Authentication Testing
- HTTP Analysis
- CSRF Analysis
- Session Handling Rules
- Burp Macros
- Intruder
- Authentication Workflow
- Response Analysis
- Web Security Testing

10. Conclusion

This assessment provided practical experience in analyzing a vulnerable authentication workflow and automating repetitive authentication tasks using Burp Suite Professional. The exercise demonstrated the importance of CSRF token handling, session management, macros, and response analysis during controlled security testing. By configuring Intruder with sequential numeric payloads and comparing HTTP responses, it was possible to identify the successful authentication condition and validate the weakness in the laboratory environment. The exercise strengthened practical skills relevant to web application security assessments and reinforced defensive recommendations for protecting authentication systems against brute-force attacks.

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